Antiferromagnetic Ising Model vs the Kitaev-Chain Qubit Hamiltonian

Seminar notes

August 12, 2026

1 The Short Version

At the sweet spot $t = \Delta$, $\mu = 0$, the Jordan-Wigner image of the Kitaev chain is

$$H_K = t \sum_{j=0}^{L-2} Y_j Y_{j+1}. \tag{1}$$

Since $t > 0$, each bond prefers opposite Y eigenvalues, so algebraically this looks like an antiferromagnetic Ising chain written in the Y basis.

The important difference is physical meaning. In the antiferromagnetic Ising model, the spins are the physical ordered variables and the two ground states are distinguished by local staggered magnetization. In the Kitaev chain, the qubits are a Jordan-Wigner encoding of fermions, and the important degeneracy is the occupation of a non-local edge fermion made from two Majorana modes. In the original fermionic language, the order is not a local staggered magnetization; it is detected by parity, edge degeneracy, and non-local string operators.

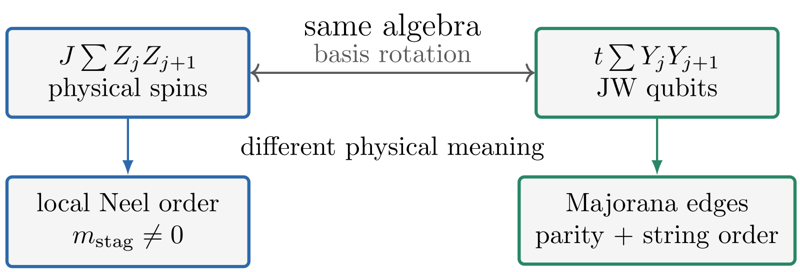

Figure 1: The central point: the sweet-spot qubit Hamiltonian has an Ising-like algebraic form, but the variables and diagnostics represent different physics.

2 Spin Variables and Pauli Operators

The classical Ising model starts with a two-valued spin variable at every site:

$$S_j = \pm 1. \tag{2}$$

For a quantum spin-$\frac{1}{2}$ degree of freedom, the local Hilbert space is two-dimensional. We choose the Z basis

$$|\uparrow\rangle_j = |0\rangle_j, \qquad |\downarrow\rangle_j = |1\rangle_j, \tag{3}$$

with

$$Z_j|\uparrow\rangle_j = +|\uparrow\rangle_j, \qquad Z_j|\downarrow\rangle_j = -|\downarrow\rangle_j. \tag{4}$$

Thus the classical value S_j is represented by the eigenvalue of the Pauli operator Z_j:

$$S_j \longleftrightarrow Z_j. \tag{5}$$

If one uses physical spin operators, they are related to Pauli matrices by

$$S_j^\alpha = \frac{\hbar}{2}\sigma_j^\alpha, \qquad \alpha = x, y, z. \tag{6}$$

In quantum-information notation we usually absorb the factors of $\hbar/2$ into the coupling constants and write Hamiltonians directly using Pauli operators X, Y, Z.

3 Antiferromagnetic Ising Hamiltonian

3.1 Classical and quantum forms

The nearest-neighbor antiferromagnetic Ising chain is

$$H_{\text{AFM}} = J\sum_{j=0}^{L-2} Z_j Z_{j+1}, \qquad J > 0. \tag{7}$$

The sign $J > 0$ means that an aligned bond has positive energy and an anti-aligned bond has negative energy.

For one bond,

$$H_{j,j+1} = J Z_j Z_{j+1}. \tag{8}$$

The four two-spin basis states are eigenstates:

State	$Z_j Z_{j+1}$ eigenvalue	Bond energy
$\lvert\uparrow\uparrow\rangle$	$+1$	$+J$
$\lvert\downarrow\downarrow\rangle$	$+1$	$+J$
$\lvert\uparrow\downarrow\rangle$	-1	$-J$
$\lvert\downarrow\uparrow\rangle$	-1	$-J$

For $J > 0$, the low-energy bond states are therefore

$$|\uparrow\downarrow\rangle, \qquad |\downarrow\uparrow\rangle. \tag{9}$$

3.2 Ground states and energies

Because all $Z_j Z_{j+1}$ terms commute, every product state in the Z basis is an eigenstate of Eq. (7). If a configuration has N_{same} aligned bonds and N_{opp} anti-aligned bonds, then

$$E = J(N_{\text{same}} - N_{\text{opp}}), \qquad N_{\text{same}} + N_{\text{opp}} = L - 1. \tag{10}$$

On an open chain, the minimum energy is obtained by making every bond anti-aligned:

$$E_0 = -J(L-1). \tag{11}$$

There are two such ground states:

$$|N_1\rangle = |\uparrow\downarrow\uparrow\downarrow\cdots\rangle, \qquad |N_2\rangle = |\downarrow\uparrow\downarrow\uparrow\cdots\rangle. \tag{12}$$

These are the two Neel states. A local defect where one bond fails to alternate changes that bond energy from $-J$ to $+J$, so one bad bond costs

$$\Delta E_{\text{bad bond}} = 2J. \tag{13}$$

These local defects are domain walls in the antiferromagnetic order.

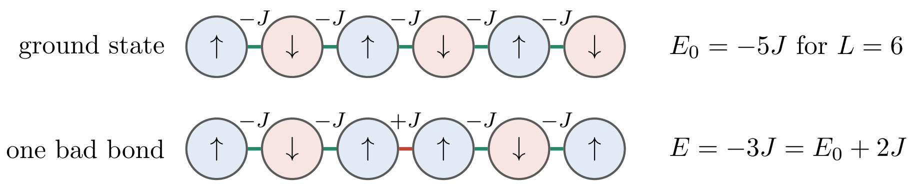

Figure 2: In the AFM Ising chain the two Neel patterns are locally visible, and a defect is a local bad bond with energy cost $2J$.

3.3 Symmetry breaking

The Ising Hamiltonian has a global spin-flip symmetry

$$U_X = \prod_{j=0}^{L-1} X_j. \tag{14}$$

This symmetry sends

$$U_X Z_j U_X^\dagger = -Z_j, \tag{15}$$

and therefore leaves every product $Z_j Z_{j+1}$ unchanged:

$$[H_{\text{AFM}}, U_X] = 0. \tag{16}$$

But U_X exchanges the two Neel states:

$$U_X|N_1\rangle = |N_2\rangle, \qquad U_X|N_2\rangle = |N_1\rangle. \tag{17}$$

On an infinite or periodic chain, a one-site translation also exchanges the two Neel patterns. Thus antiferromagnetic order doubles the unit cell: translation by one site is broken down to translation by two sites. On an open chain, translation is not an exact symmetry, but the same staggered pattern is visible locally in the bulk.

The local order parameter is the staggered magnetization

$$m_{\text{stag}} = \frac{1}{L}\sum_{j=0}^{L-1}(-1)^j Z_j. \tag{18}$$

It has opposite signs in the two ground states:

$$\langle N_1|m_{\text{stag}}|N_1\rangle \approx +1, \qquad \langle N_2|m_{\text{stag}}|N_2\rangle \approx -1, \tag{19}$$

up to the convention for which sublattice is labelled even.

This is conventional symmetry breaking: the Hamiltonian has the symmetry, but an ordered ground state chooses one of the two symmetry-related patterns. In a finite system one can form symmetric and antisymmetric cat states, but in the thermodynamic limit an arbitrarily small staggered field selects one Neel pattern. The two phases can be distinguished by a local measurement of m_{stag}.

3.4 Adding a transverse field

The transverse-field antiferromagnetic Ising model is

$$H_{\mathrm{TF-AFM}} = J \sum_{j=0}^{L-2} Z_j Z_{j+1} - h \sum_{j=0}^{L-1} X_j, \qquad J > 0. \tag{20}$$

The transverse-field term flips spins:

$$X_j |\uparrow\rangle_j = |\downarrow\rangle_j, \qquad X_j |\downarrow\rangle_j = |\uparrow\rangle_j. \tag{21}$$

Therefore the classical product states are no longer exact eigenstates when $h \neq 0$. For small h/J, the system remains antiferromagnetically ordered; for large h/J, it becomes a paramagnet polarized along X. The transition is a symmetry-breaking transition between a staggered ordered phase and a disordered phase.

4 Kitaev Chain and Its Qubit Hamiltonian

4.1 Fermionic model

The open Kitaev chain is

$$H_K = -\mu \sum_{j=0}^{L-1} c_j^\dagger c_j - t \sum_{j=0}^{L-2} (c_j^\dagger c_{j+1} + \mathrm{h.c.}) + \Delta \sum_{j=0}^{L-2} (c_j^\dagger c_{j+1}^\dagger + \mathrm{h.c.}). \tag{22}$$

This is not originally a spin model. It is a superconducting fermion model: t moves fermions, Δ creates or destroys fermion pairs, and μ controls occupation. Fermion number is not conserved because of the pairing term, but fermion parity is conserved:

$$P = (-1)^{\sum_j c_j^\dagger c_j}. \tag{23}$$

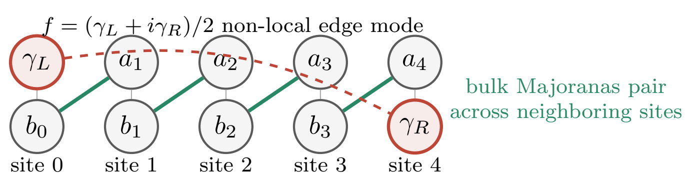

Figure 3: At the topological sweet spot, the bulk Majoranas pair locally while one Majorana remains at each end. The two edge Majoranas form a non-local fermion whose occupation labels the near-degenerate parity sectors.

4.2 Jordan-Wigner qubit form

Under the Jordan-Wigner mapping, the Hamiltonian becomes

$$H_K = -\frac{\mu}{2}\sum_{j=0}^{L-1} Z_j + \frac{t-\Delta}{2}\sum_{j=0}^{L-2} X_j X_{j+1} + \frac{t+\Delta}{2}\sum_{j=0}^{L-2} Y_j Y_{j+1}. \tag{24}$$

The qubit Z_j has a direct fermionic meaning:

$$n_j = c_j^\dagger c_j = \frac{I - Z_j}{2}. \tag{25}$$

The total fermion parity is

$$P = \prod_{j=0}^{L-1} Z_j. \tag{26}$$

The X_j and Y_j operators are not simply local fermion observables; under the inverse Jordan-Wigner map they include parity strings. This is why local spin order in the qubit language can correspond to non-local string order in the fermion language.

5 Sweet Spot: Why It Looks Like an AFM Ising Chain

At $t = \Delta$, the XX term cancels:

$$H_K(t = \Delta) = -\frac{\mu}{2}\sum_j Z_j + t\sum_{j=0}^{L-2} Y_j Y_{j+1}. \tag{27}$$

At $\mu = 0$,

$$H_K(t = \Delta, \mu = 0) = t\sum_{j=0}^{L-2} Y_j Y_{j+1}. \tag{28}$$

This is algebraically the same structure as an antiferromagnetic Ising chain, but in the Y basis rather than the Z basis.

If μ is turned back on, $-\mu\sum_j Z_j/2$ does not commute with the $Y_j Y_{j+1}$ ordering term. After a spin-axis rotation that sends Y to Z, this term plays the algebraic role of a transverse field. This is why the qubit Hamiltonian resembles a transverse-field Ising model near the sweet spot. The physical meaning is still different: μ is a chemical potential for fermions, and the transition at $|\mu| = 2t$ is the closing and reopening of a topological superconducting gap.

Let $|+y\rangle$ and $|-y\rangle$ be eigenstates of Y:

$$Y|+y\rangle = +|+y\rangle, \qquad Y|-y\rangle = -|-y\rangle. \tag{29}$$

In the occupation/computational basis,

$$|+y\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \qquad |-y\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}. \tag{30}$$

So a Y-basis Neel state is already a coherent superposition of many fermion occupation configurations. It should not be read as a classical pattern of particles sitting on alternating sites.

For one bond,

$$tY_j Y_{j+1} \tag{31}$$

has energy $+t$ when neighboring Y eigenvalues are equal and energy $-t$ when they are opposite. Thus, for $t > 0$, the two qubit-spin ground states of Eq. (28) are

$$|\widetilde{N}_1\rangle = |+y, -y, +y, -y, \ldots\rangle, \qquad |\widetilde{N}_2\rangle = |-y, +y, -y, +y, \ldots\rangle, \tag{32}$$

with energy

$$E_0 = -t(L-1). \tag{33}$$

The parity operator $P = \prod_j Z_j$ maps these two states into each other:

$$P|\widetilde{N}_1\rangle = |\widetilde{N}_2\rangle, \qquad P|\widetilde{N}_2\rangle = |\widetilde{N}_1\rangle. \tag{34}$$

Since $[H_K, P] = 0$, one may choose definite-parity ground states instead:

$$|\psi_\pm\rangle = \frac{1}{\sqrt{2}}\left(|\widetilde{N}_1\rangle \pm |\widetilde{N}_2\rangle\right), \qquad P|\psi_\pm\rangle = \pm|\psi_\pm\rangle. \tag{35}$$

This is the first place where the interpretation changes: the same degenerate subspace can be written as two Y-ordered spin states or as two fermion-parity states. The Kitaev-chain interpretation uses the parity states and the edge Majorana mode that connects them.

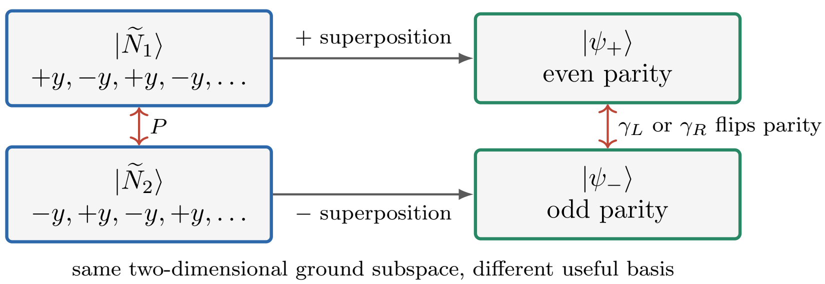

Figure 4: The pure YY Hamiltonian has two Y-Neel product states, but the Kitaev interpretation reorganizes this subspace into definite fermion-parity states connected by edge Majorana operators.

So, as a spin Hamiltonian, the sweet-spot Kitaev qubit Hamiltonian is an antiferromagnetic Ising chain after rotating the spin axes from Y to Z. That statement is correct but incomplete.

6 Where the Physics Diverges

6.1 Different degrees of freedom

In the AFM Ising model, the spins are the physical ordered variables. Measuring Z_j locally tells us whether the system chose one staggered pattern or the other.

In the Kitaev problem, the qubits encode fermions. The Z_j operator measures local occupation parity, but X_j and Y_j contain Jordan-Wigner strings in fermion language. Therefore the Y-Neel order in Eq. (32) is not an ordinary local magnetic order of the original fermions. It is the spin-chain image of superconducting/topological order.

6.2 Different meaning of degeneracy

For the AFM Ising chain, the two ground states are two locally distinguishable patterns:

$$|\uparrow\downarrow\uparrow\downarrow\cdots\rangle \quad \text{versus} \quad |\downarrow\uparrow\downarrow\uparrow\cdots\rangle. \tag{36}$$

The degeneracy is tied to symmetry breaking. A local staggered field can select one pattern.

For the open Kitaev chain in the topological phase, the degeneracy is tied to Majorana edge modes. The two edge Majoranas combine into a non-local fermionic mode

$$f = \frac{1}{2}(\gamma_L + i\gamma_R). \tag{37}$$

The two low-energy states differ by whether this non-local fermion is empty or occupied:

$$n_f = f^\dagger f = 0 \quad \text{or} \quad 1. \tag{38}$$

Changing n_f flips total fermion parity. A local parity-preserving fermion operator cannot directly change this edge occupation, because that would require coupling the two ends of the chain. In a finite chain away from the exact sweet spot, the splitting between the two parity sectors is exponentially small in system size:

$$|E_0^{(+)} - E_0^{(-)}| \sim e^{-L/\xi}. \tag{39}$$

This is the Majorana hybridization splitting, not the energy cost of a local spin-domain-wall defect.

6.3 Different symmetry story

The AFM Ising Hamiltonian has a global spin-flip symmetry, and the ordered phase breaks that symmetry by selecting one Neel pattern. When translation is a symmetry, the same order also breaks one-site translation down to two-site translation. The order parameter is local:

$$m_{\text{stag}} = \frac{1}{L}\sum_j (-1)^j Z_j. \tag{40}$$

The Kitaev chain has conserved fermion parity:

$$[H_K, P] = 0. \tag{41}$$

The topological phase does not require a local order parameter that breaks fermion parity. In fact, operators corresponding to a single Majorana edge mode flip parity, so their expectation value vanishes in a parity eigenstate. The robust diagnostic is parity-preserving and non-local, for example

$$O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}. \tag{42}$$

For $L = 4$,

$$O_{\text{edge}} = X_0 Z_1 Z_2 X_3. \tag{43}$$

This is why Week 5 and Week 6 focus on string/correlation observables rather than local magnetization.

6.4 Different protection mechanism

AFM order is stable because a local domain wall costs finite energy. However, the two ordered patterns are locally distinguishable, and a local staggered field directly favors one over the other.

The Kitaev topological degeneracy is protected differently. The two Majoranas live at opposite ends of an open chain, so a local parity-preserving perturbation cannot easily couple them. The splitting is controlled by the overlap of edge-mode wavefunctions and is exponentially small in L. This is a non-local protection mechanism, not ordinary local symmetry breaking.

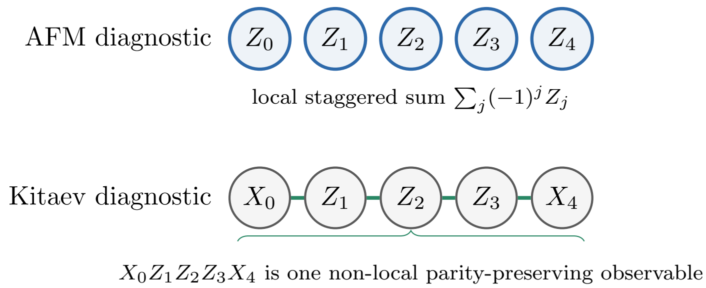

Figure 5: The Ising order parameter is a sum of local measurements. The Kitaev edge diagnostic is a single non-local string that ties the two ends together while preserving total fermion parity.

7 Comparison Table

Feature	AFM Ising chain	Kitaev-chain qubit Hamiltonian
Simple Hamiltonian	$J \sum Z_j Z_{j+1}$, $J > 0$	$t \sum Y_j Y_{j+1}$ at $t = \Delta, \mu = 0$
Low-energy bond	opposite Z eigenvalues	opposite Y eigenvalues
Ground states	two Z-basis Neel states	two Y-basis Neel-like qubit states at the sweet spot
Physical variables	spins are the ordered variables	qubits encode fermions via Jordan-Wigner
Degeneracy source	broken discrete spin symmetry	Majorana edge mode / fermion parity sectors
Order parameter	local staggered magnetization	non-local string / edge correlation in fermion language
Excitations	local domain walls	bulk quasiparticles plus edge-mode parity splitting
Protection	local energy cost for bad bonds	exponentially weak edge-edge coupling in an open chain

8 Takeaway

The sweet-spot Kitaev qubit Hamiltonian and an AFM Ising chain can have the same algebraic form after a spin-basis rotation:

$$J \sum Z_j Z_{j+1} \quad \leftrightarrow \quad t \sum Y_j Y_{j+1}. \tag{44}$$

But they answer different physical questions. The AFM Ising model describes local staggered spin order and symmetry breaking. The Kitaev chain describes a topological superconducting phase whose qubit Hamiltonian only looks like a spin model after Jordan-Wigner encoding. Its key signatures are conserved fermion parity, Majorana edge modes, exponentially small parity splitting, and non-local string correlations.